

localStorage support on browsers/platforms

This table lists the minimum version of each browser required to support localStorage:

IE	Firefox	Safari	Chrome	Opera	iPhone	Android
8.0+	3.5+	4.0+	4.0+	10.5+	2.0+	2.0+

Operations on localStorage

The methods of *localStorage* include *setItem()*, *getItem()*, *removeItem()* and *clear()*. Here is sample code to add a record to *localStorage* using *setItem()*:

```
// max limit reached exception
try {
    localStorage.setItem(itemId, "Sample Value");
} catch (e) {
    if (e == QUOTA_EXCEEDED_ERR) {
        alert('Quota exceeded!');
    }
}
```

We will use *getItem()* and *removeItem()* in the Web app that we will develop. Note that *localStorage.clear()* removes all entries from the database, so use it with caution!

Debugging localStorage

The quick and easy way to view the *localStorage* database is in Chrome, as part of Developer Tools. Select the wrench menu at the top-right corner, go to *Tools -> Developer Tools*. Click the *Resources* tab, select *Local Storage* and then *Local Files*. The right pane should display key-value pairs (Figure 1), if the database has data. For debugging in Firefox, you need the Firebug extension.

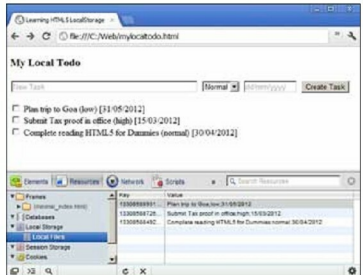


Figure 1: Database view



Figure 2: Task listing



Figure 3: Tool-tip notification for mandatory entry

MyLocalTodo: An example of the use of localStorage

Let's try implementing a *My Local Todo* Web application, which can work offline, with minimal code. Besides *localStorage*, we will also use HTML5 features such as *placeholder* and *required*.

Application UI

In this simple to-do application, the first (text) field 'New Task' is the task description; the second is task priority, and third the due date. Only the first is mandatory. If the user clicks 'Create Task' without entering it, the browser shows a tool-tip below the text field (Figure 3). The default task priority is 'Normal' and the due date is today.

```
1: <body onLoad="browserCheck();getTasks();">
2:   <h3>My Local Todo</h3>
3:   <form name="todoForm" onSubmit="addTask();">
4:     <input type="text" size="50" name="task" placeholder="New Task" value="" autofocus required>
```

```
5:     <select id="priority">
6:       <option value="high">High</option>
7:       <option value="normal" selected=Normal</option>
8:       <option value="low">Low</option>
9:     </select>
10:    <input type="text" size="10" name="duedate"
    placeholder="dd/mm/yyyy">
11:    <input type="submit" value="Create Task">
12:  </form>
13:  <p id="theTaskList">Nothing much to do!</p>
14: </body>
```

Line 1: We check if the browser supports localStorage. If not, display an alert. We also display tasks already present in localStorage.

Line 3-12: Form with fields to accept task text, priority and